



ENERGY
CASE
CHALLENGE

MARCH 2025

Strategic Report

FOR INA SOLAR'S ENTRY INTO THE
ENERGY STORAGE MARKET.

PRESENTED BY

Diva Agrawal
Jay Lunawat

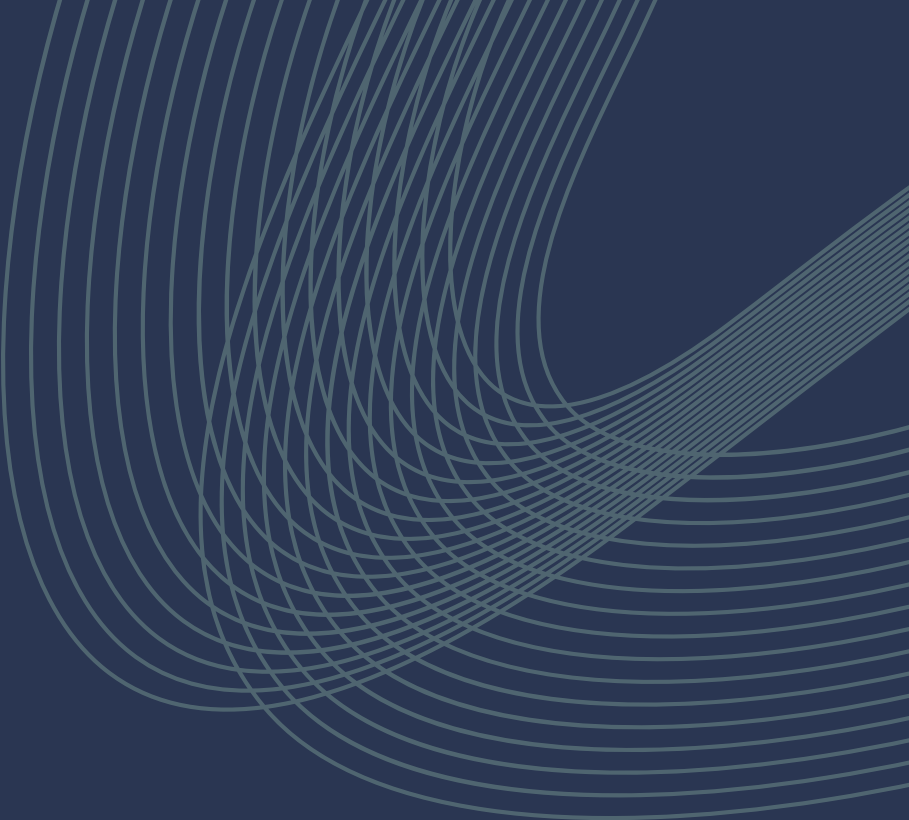


TABLE OF CONTENTS

Synopsis	3
Company Summary	4
Battery Technology Selection	5
Market Segment Analysis	7
Technical Compatibility and Synergy Analysis	8
Economic Feasibility	9
Strategic Recommendations	10

Synopsis

INA Solar, a leader in India's solar energy sector, is strategically positioned to enter the rapidly growing battery energy storage market. This report evaluates **Lithium Iron Phosphate (LFP) batteries** as the optimal technology choice due to their safety, longevity, and cost-effectiveness. The recommended market segments are:

- Utility-scale Battery Energy Storage Systems (BESS) (Primary focus)
- Commercial & Industrial (C&I) Rooftop Solar-Plus-Storage (Secondary focus)

The roadmap leverages INA Solar's existing manufacturing capabilities, EPC expertise, and strong market presence to ensure a seamless entry into battery storage. A phased investment approach with an estimated **5-7 year payback period** is proposed, supported by government incentives and growing market demand.

COMPANY SUMMARY

INA Solar has established itself as a leader in solar PV manufacturing, with:

- 1,000 MW production capacity (expanding to 4,000 MW).
- 10,000+ clients, 600+ projects, and 700+ channel partners.
- Strong government collaborations (PM Kusum Yojana, SECI, JJM).

With India's energy storage market projected to grow at 22% CAGR, INA Solar must capitalize on this opportunity by aligning battery storage with its solar business.



BATTERY TECHNOLOGY SELECTION

Technology Options:

Technology	Advantages	Disadvantages	Suitability*
LFP (LiFePO_4)	• High safety. • Long lifespan (3000 -5000 cycles). • Cost-effective.	• Lower energy density.	• Best for stationary storage. • High temperatures.
NMC (Li-NMC)	• Higher energy density. • Better performance in varied temps.	• Higher cost. • Safety risk.	• More suited for EVs than stationary storage.
Sodium-ion	• Low-cost materials. • Good thermal stability.	• Lower energy density. • Early-stage tech.	• Future potential, but not yet mature.
Solid-state	• Ultra-safe. • High energy density • Fast charging.	• Extremely high cost. • In R&D phase.	• Long-term option, not viable now.

*for India.

BATTERY TECHNOLOGY SELECTION

Evaluation Criteria:

- Safety (critical for Indian climate).
- Cycle life & degradation (longevity matters for Return of Investment).
- Cost per kWh (LFP is most economical).
- Supply chain maturity (LFP has established vendors).

Recommended Technology: LFP Batteries

- Best fit for India's high-temperature conditions.
- Lower fire risk compared to NMC.
- Compatible with solar integration (existing INA Solar expertise).

MARKET SEGMENT ANALYSIS

Target Segments:

Primary Segment: Utility-Scale BESS (60% Focus)

- **Market Drivers:**
 - Government tenders (SECI*, state DISCOMs).
 - Renewable energy integration mandates.
- **Technical Specs:**
 - 10MWh+ capacity, 4-8 hour discharge.
 - Grid-stabilisation capabilities.

Secondary Segment: C&I Rooftop Solar+Storage (40% Focus)

- **Market Drivers:**
 - Rising demand for backup power.
 - High electricity costs for industries.
- **Technical Specs:**
 - 50kWh-2MWh systems.
 - Seamless solar coupling.

*Solar Energy Corporation of India Ltd.

TECHNICAL COMPATIBILITY & SYNERGY ANALYSIS

Manufacturing & EPC Synergies

- Existing battery assembly lines (PCU production) can be adapted.
- EPC expertise in large-scale solar projects translates well to BESS.

Supply Chain & Customer Base

- 700+ channel partners can distribute C&I storage solutions.
- Existing relationships with DISCOMs aid utility-scale projects.

ECONOMIC FEASIBILITY

Capital Investment

Component	Estimated Cost (₹ Crore)
• 250 MWh Pilot Line	300-400
• 1 GWh Full Scale	1,200-1,500

Revenue Potential

Segment	Price Range (₹ Crore/MWh)	Margins
• Utility-Scale	5.5-6.5	15-20%
• C&I*	7-8	25-30%

Payback Period: 5-7 years

*Commercial and Industrial

STRATEGIC RECOMMENDATIONS

- **Short-Term (0-18 Months)**
 - Set up 250 MWh LFP pilot production.
 - Bid for SECI/state storage tenders.
 - Launch C&I solar+storage packages.
- **Medium-Term (18-36 Months)**
 - Scale to 1 GWh production.
 - Expand into microgrids & industrial storage.
- **Long-Term (36-60 Months)**
 - Target 5 GWh capacity.
 - Explore sodium-ion/solid-state for future.

INA Solar's entry into battery storage with LFP technology and a focus on utility-scale and C&I segments is a strategically sound move. By leveraging existing strengths and adopting a phased investment approach, INA Solar can establish itself as a key player in India's energy storage market.

